Yongchun Li - Week 3 Report (1378156)

1. Tasks Completed This Week

See PR: https://github.com/SWEN90017-2025-SM1/OAS-web-based-oral-assessments/pull/284

1. **Epic 4 Course Management Implementation (6h)**
    - Analyzed Epic 4.1 user stories and designed comprehensive Course management system
    - Simplified academic structure design by embedding academic year and semester directly into Subject entity to align with frontend expectations
    - Implemented complete CRUD operations for Course entity with proper validation and error handling
    - Participated in team meetings to discuss and coordinate Assignment and Rubric functionality implementation
2. **Database Schema Design & Optimization (2h)**
    - Updated `subject` table with embedded academic year/semester fields for frontend compatibility
    - Collaborated on `assignment` table design with comprehensive rubric support via JSON structure
    - Implemented optimized SQL queries for dashboard statistics using aggregation (avoiding N+1 query problems)
    - Added proper indexing and unique constraints for data integrity
3. **API Architecture & Documentation (3h)**
    - Created standardized DTOs for Course operations with comprehensive validation
    - Refactored Course controller to use `Result<T>` pattern instead of generic `Map<String, Object>` for better type safety
    - Generated AI-friendly API documentation with complete examples and validation rules
    - Added comprehensive Swagger annotations across Course management related code
    - Assisted team members with Assignment API design and documentation standards
4. **Team Collaboration & Advanced Features (1h)**
    - Participated in design discussions for flexible JSON-based rubric structure supporting hierarchical criteria (Epic 4.4-4.6)
    - Contributed to performance level configuration system design with customizable points and descriptions
    - Collaborated on extensible comment library system integrated into rubric structure

Time Allocation: Approximately 12 hours total

- (6 hours Course implementation, 2 hours database design, 3 hours API documentation, 1 hour team collaboration)

2. Key Issues Encountered

- **Initial Design Complexity:** The initial separate `academic_year` and `semester` tables approach proved overly complex compared to frontend needs, leading to a major refactor for simplification.
- **Performance Optimization:** Identified and resolved N+1 query issues in statistics endpoints by implementing SQL-level aggregation.
- **Code Style Consistency:** Required multiple iterations to align controller return types with project standards, moving from `ResponseEntity` to standardized `Result<T>` pattern.

3. Next Week's Plan

- Continue development of remaining Epic components (Epic 5: Booking and Scheduling System)
- Implement email notification system integration for course and assignment updates
- Support final presentation preparation and system integration testing
- Collaborate on frontend-backend API integration and testing

4. Technical Achievements

Backend Implementation Highlights:

- **Course CRUD APIs:** Implemented complete Course management with pagination, filtering, and statistics
- **Database Optimization:** Single-query statistics using SQL aggregation instead of N+1 patterns
- **Cascade Operations:** Intelligent cascade deletion for courses with associated assignments
- **Team Coordination:** Contributed to overall Epic 4 architecture and Assignment/Rubric system design

Code Quality Improvements:

- **Type Safety:** Eliminated generic `Map<String, Object>` returns in favor of structured DTOs
- **Documentation:** Comprehensive Swagger annotations for Course management APIs
- **Validation:** Complete request validation with proper error handling and business error codes
- **Performance:** Optimized database queries and efficient data structures

API Coverage:

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Course Management: 6 endpoints (CRUD + stats + filters)
Supporting DTOs: 8 request/response objects with full validation
Team Coordination: Assisted with Assignment management design
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Epic 4.1 (Course Management) has been successfully implemented and documented, with collaborative contributions to Epic 4.2-4.6 (Assignment & Rubric systems), providing a solid foundation for the OAS course management system.